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SECTION : Pro. Amit Sharma

SI-SG_AIMLCZC418 - Assignment 1

Section : Prof. Amit Sharma

Q-1

Given Data,

Recorded travel time (in minutes) for $n = 10$ days

42, 45, 47, 44, 46, 48, 43, 45, 47, 85

Ordered data (Ascending order)

42, 43, 44, 45, 45, 46, 47, 47, 48, 85

sample size $n = 10$

a) Mean ($\bar{x}$) & Median ($\tilde{x}$)

1) Mean ($\bar{x}$)

$$\bar{x} = \frac{\sum_{i=1}^{n} x_i}{n}$$

$$\sum_{i=1}^{10} x_i = 492$$

$$\bar{x} = \frac{492}{10}$$

$$\boxed{\bar{x} = 49.2 \text{ minutes}}$$

2) Median ($\tilde{x}$)

$$\tilde{x} = \frac{x_{(n/2)} + x_{(n/2+1)}}{2}$$

$n/2 = 10/2 = 5^{th}$ value $= 45$

$n/2 + 1 = 6^{th}$ value $= 46$

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$$\tilde{x} = \frac{45+46}{2}$$

$$\tilde{x} = 45.5 \text{ minutes}$$

b) Outlies detection using the IQR Method (1.5 x IQR Rule)

→ finding Q_1 & Q_3

Lower Half { 42, 43, 44, 45, 45 }

Upper Half { 46, 47, 47, 48, 85 }

$Q_1 = 44$ minutes

$Q_3 = 47$ minutes

→ Interquartile Range (IQR)

$$IQR = Q_3 - Q_1$$

$$IQR = 43 - 44$$

$$IQR = 3 \text{ minutes}$$

- Outlier limits

Lower Limit = $Q_1 - 1.5 \times IQR$

Upper Limit = $Q_3 + 1.5 \times IQR$

Lower limit = $44 - 4.5$ = 39.5 minutes

Upper limit = $47 - 4.5$ = 51.5 minutes

Minimum value 42 > 39.5 (No lower outlier)

Maximum Value 85 > 51.5 (Upper Outlier)

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e)

→ The Median gives a better representation of the typical travel time

→ Justification :-

1) sensitivy vs resistance : Mean is highly sensitive to extreme values (outliers)

2) Robustness of the Median :-

- median is a resistant measure of central tendency at 45.5 minutes it sits directly in the center of the dense cluster of daily observation accuracy reflecting the student's typical daily commute

Q-2

Given data,

Red Balls : $n(R) = 5$

Blue balls : $n(B) = 4$

Green balls : $n(G) = 3$

total number of Balls :

$$N = n(R) + n(B) + n(G)$$
$$= 5 + 4 + 3$$
$$N = 12$$

a) Probability that the selected ball is red or blue

$$P(R \cup B) = P(R) + P(B) = \frac{n(R) + n(B)}{N}$$

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$$P(R \cup B) = \frac{5+4}{12}$$

$$= 0.75$$

b) Probality selected Ball is not green

$$P(G^c) = 1 - P(G) = 1 - \frac{n(G)}{N}$$

$$= 1 - \frac{3}{12}$$

$$= 0.75$$

c) Manually Exclusive Determination & Justification

- yes, the events are mutually excusive
- Reason: only one ball is drawn at random. A single ball cannot be simultaneously red & blue meaning the two outcomes cannot happens at the same time

$$R \cap B = \phi \Rightarrow P(R \cap B) = 0$$

Q-3

given Data,

A : component Produced by Machine A
B : component Produced by Machine B
D : Component is defective

Prior Probability

$P(A) = 70\% = 0.70$

$P(B) = 30\% = 0.30$

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Conditional Probability

$P(D|A) = 0.02$

$P(D|B) = 0.05$

a) Overall Probability that the selected component is defective

$P(D) = P(A) \cdot P(D|A) + P(B) \cdot P(D|B)$

$P(D) = (0.70 \times 0.02) + (0.30 \times 0.05)$

$$\boxed{P(D) = 0.029}$$

b) Probability that defective component was produced by Machine

$$P(B|D) = \frac{P(B) \cdot P(D|B)}{P(D)}$$

$$P(B|D) = \frac{0.30 \times 0.05}{0.029}$$

$$\boxed{P(B|D) = \frac{15}{29} \approx 0.5172}$$

Q-4

Given Probability Distribution

X	0	1	2	3
P(X)	0.40	0.30	0.20	0.10

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a) Verify Valid Probability Distribution

1. non negativity condition:

$$0 \leq P(X = x) \leq 1 \quad \text{for all } x$$

Verification:

$P(0) = 0.40 \in [0,1]$

$P(1) = 0.30 \in [0,1]$

$P(2) = 0.20 \in [0,1]$

$P(3) = 0.10 \in [0,1]$

2. Total Probability condition

$$\sum_{\text{all } x} P(X = x) = 1$$

Verification

$$\sum P(x) = 0.40 + 0.30 + 0.20 + 0.10 = 1.00$$

- Both conditions are satisfied so the given values form a valid probability distribution

b) Expected Number of Item Purchased, $E(X)$

$$E(X) = \sum x \cdot P(X = x)$$

$$E(X) = (0 \times 0.40) + (1 \times 0.30) + (2 \times 0.20) + (3 \times 0.10)$$

$$\boxed{E(X) = 1.00}$$

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c) Probability of Purchasing Two or More Items

$$P(X \geq 2) = P(X=2) + P(X=3)$$

$$P(X \geq 2) = 0.20 + 0.10 = 0.30$$